Q1. a) $t^x \cdot e^{-t} = e^{-t+x\ln t}$

$\therefore$ Let $t = sx$, $\frac{ds}{dt} = \frac{1}{x}$ $\Rightarrow$ $dt = x\,ds$

$$\therefore \Gamma(x+1) = \int_0^\infty t^x e^{-t}\,dt = x^{x+1}\int_0^\infty s^x e^{-sx}\,ds = x^{x+1}\int_0^\infty e^{-x(s-\ln s)}\,ds$$

b) Let $s-\ln s \equiv 1+\tau^2$

```
In[92]:= (*1. 定义偏移后的函数，使其在 s=1 时等于 0*) f = s - Log[s] - 1
                                                     对数

(*2. 在 s=1 处展开。注意：为了得到 C2，s 至少要展开到 6 阶*)
sExp = Series[f, {s, 1, 8}]
       级数
(*3. 关键：取平方根。使用 PowerExpand 确保 Mathematica 处理符号正确*)
(*这样得到的 tauSeries 在 s=1 处的一阶项系数不为 0*)
tauSeries = Sqrt[sExp] // PowerExpand
            平方根          幂展开
(*4. 此时可以安全执行 InverseSeries，得到 s 关于 tau 的级数*)
sMinus1OfTau = InverseSeries[tauSeries, tau]
               反函数级数
(*5. 计算 ds/dtau 并展开*)
dsDtau = D[sMinus1OfTau, tau] // Series[#, {tau, 0, 4}] & // Normal
         偏导                    级数                          转换为普通表

Print["ds/dtau 的级数展开为: ", dsDtau]
打印
```

Out[92]= $-1 + s - \text{Log}[s]$

Out[93]= $\frac{1}{2}(s-1)^2 - \frac{1}{3}(s-1)^3 + \frac{1}{4}(s-1)^4 - \frac{1}{5}(s-1)^5 + \frac{1}{6}(s-1)^6 - \frac{1}{7}(s-1)^7 + \frac{1}{8}(s-1)^8 + O[s-1]^9$

Out[94]= $\frac{s-1}{\sqrt{2}} - \frac{(s-1)^2}{3\sqrt{2}} + \frac{7(s-1)^3}{36\sqrt{2}} - \frac{73(s-1)^4}{540\sqrt{2}} + \frac{1331(s-1)^5}{12960\sqrt{2}} - \frac{22409(s-1)^6}{272160\sqrt{2}} + O[s-1]^7$

Out[95]= $1 + \sqrt{2}\,\text{tau} + \frac{2\,\text{tau}^2}{3} + \frac{\text{tau}^3}{9\sqrt{2}} - \frac{2\,\text{tau}^4}{135} + \frac{\text{tau}^5}{540\sqrt{2}} + \frac{4\,\text{tau}^6}{8505} + O[\text{tau}]^7$

Out[96]= $\sqrt{2} + \frac{4\,\text{tau}}{3} + \frac{\text{tau}^2}{3\sqrt{2}} - \frac{8\,\text{tau}^3}{135} + \frac{\text{tau}^4}{108\sqrt{2}}$

ds/dtau 的级数展开为: $\sqrt{2} + \frac{4\,\text{tau}}{3} + \frac{\text{tau}^2}{3\sqrt{2}} - \frac{8\,\text{tau}^3}{135} + \frac{\text{tau}^4}{108\sqrt{2}}$

c) $\Gamma(x+1) = x^{x+1}\int_0^{\infty} e^{-x(s-\ln s)}\,ds = x^{x+1}\int_0^{\infty} e^{-x(1+\tau^2)}\,\frac{ds}{d\tau}\cdot d\tau$

$= x^{x+1}\cdot e^{-x}\int_{-\infty}^{\infty} e^{-x\tau^2}\left[\sqrt{2}+\frac{4}{3}\tau+\frac{1}{3\sqrt{2}}\tau^2+\frac{8}{135}\tau^3+\frac{1}{108\sqrt{2}}\tau^4\right]d\tau$

Consider $\tau$ 、 $\tau^3$ makes $\int_{-\infty}^{\infty}\tau^{2k+1}e^{-x\tau^2}d\tau=0$ , $k\in\mathbb{Z}^+$

$\Gamma(x+1) = x^{x+1}\cdot e^{-x}\left[\int_{-\infty}^{\infty}\sqrt{2}\cdot e^{-x\tau^2}d\tau+\int_{-\infty}^{\infty}\frac{1}{3\sqrt{2}}\tau^2 e^{-x\tau^2}d\tau+\int_{-\infty}^{\infty}\frac{1}{108\sqrt{2}}\tau^4 e^{-x\tau^2}d\tau\right]$

$\int_{-\infty}^{\infty}\sqrt{2}e^{-x\tau^2}d\tau = \sqrt{\frac{2\pi}{x}}$ , $C_0 = 1$

$\int_{-\infty}^{\infty}\frac{1}{3\sqrt{2}}\tau^2 e^{-x\tau^2}d\tau = \frac{1}{3\sqrt{2}}\left[\int_{-\infty}^{\infty}\frac{1}{2x}e^{-x\tau^2}d\tau-\underbrace{\frac{1}{2x}\tau e^{-x\tau^2}\Big|_{-\infty}^{\infty}}_{0}\right]=\frac{1}{6\sqrt{2}x}\sqrt{\frac{\pi}{x}}$ , $C_1=\frac{1}{12}$

$\int_{-\infty}^{\infty}\frac{1}{108\sqrt{2}}\tau^4 e^{-x\tau^2}d\tau = \int_{-\infty}^{\infty}\frac{3}{4x^2}\cdot\frac{1}{108\sqrt{2}}e^{-x\tau^2}d\tau = \sqrt{\frac{\pi}{x}}\cdot\frac{1}{x^2}\cdot\frac{1}{4\times 36\times\sqrt{2}}$ , $C_2=\frac{1}{288}$

$\therefore n! \sim n^n\cdot e^{-n}\sqrt{2\pi n}\left(1+\frac{1}{12}\cdot\frac{1}{n}+\frac{1}{288}\cdot\frac{1}{n^2}+\cdots\right)$

Q2.

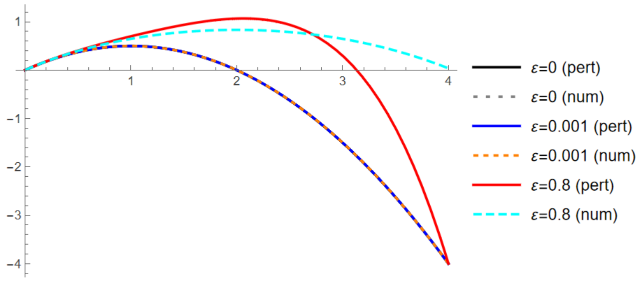
a) The results up to $O(\varepsilon^1)$ is: $y = \frac{x}{x_c}$, $\tau = \frac{t}{t_c}$; $x_c = \frac{V_0^2}{g}$, $t_c = \frac{V_0}{g}$.

$$y = -\frac{\tau^2}{2} + \tau + \varepsilon\left(-\frac{1}{12}\tau^4 + \frac{1}{3}\tau^3\right) + O(\varepsilon^2)$$

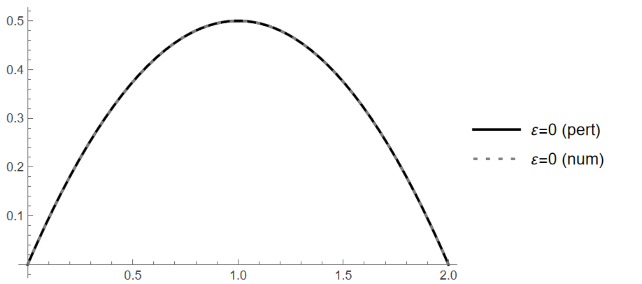
```
In[298]:= Clear[y, yNum, τ, ε]
          清除
y[τ_, ε_] := τ - τ^2/2 + ε * (τ^3/3 - τ^4/12);
sol[ε_] := NDSolve[{yNum''[τ] == -1/(1 + ε yNum[τ])^2, yNum[0] == 0, yNum'[0] == 1}, yNum, {τ, 0, 2}];
           数值求解微分方程组
Plot[{y[τ, 0], Evaluate[yNum[τ] /. sol[0]], y[τ, 0.001], Evaluate[yNum[τ] /. sol[0.001]], y[τ, 0.8], Evaluate[yNum[τ] /. sol[0.8]]}, {τ, 0, 4},
绘图             计算                                         计算                                      计算
 PlotStyle → {{Black, Thick}, {Gray, Thick, Dashing[{0.01, 0.02}]}, {Blue, Thick}, {Orange, Thick, Dashed}, {Red, Thick}, {Cyan, Thick, Dashing[{0.02, 0.01}]} },
 绘图样式       黑色    粗      灰色   粗     虚线线段配置              蓝色   粗      橙色    粗     虚线     红色  粗      蓝绿色 粗     虚线线段配置
 PlotLegends → {"ε=0 (pert)", "ε=0 (num)", "ε=0.001 (pert)", "ε=0.001 (num)", "ε=0.8 (pert)", "ε=0.8 (num)"}]
 绘图的图例

(*对比 ε = 0 的情况*)
Plot[{y[τ, 0], Evaluate[yNum[τ] /. sol[0]]}, {τ, 0, 2}, PlotStyle → {{Black, Thick}, {Gray, Thick, Dashing[{0.01, 0.02}]} }, PlotLegends → {"ε=0 (pert)", "ε=0 (num)"}]
绘图             计算                                   绘图样式       黑色    粗      灰色   粗     虚线线段配置               绘图的图例
(*对比 ε = 0 在末端的区别*)
Plot[{y[τ, 0.001], y[τ, 0]}, {τ, 1.99, 2.01}, PlotStyle → {{Black, Thick}, {Gray, Thick, Dashing[{0.01, 0.02}]} }, PlotLegends → {"ε=0.001 (pert)", "ε=0 (pert)"}]
绘图                                          绘图样式       黑色    粗      灰色   粗     虚线线段配置               绘图的图例
```

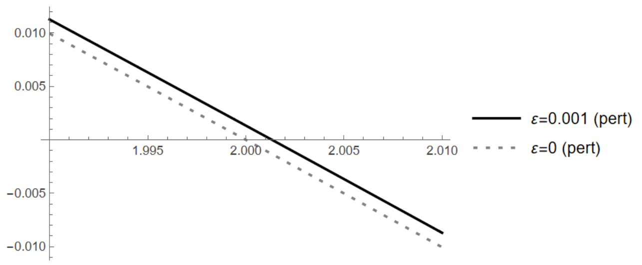
Out[301]=




Out[302]=




Out[303]=




As the results:
When $\varepsilon = 0$, the two methods show the same plots.
When $\varepsilon = 0.001$, they are almost the same.
When $\varepsilon = 0.8$, the results are quietly different.

I tried locally magnificant the results of purturbation method when $\varepsilon = 0$ and $0.001$. And find $\tau^* > 2.0$.

c) Try $\tau^* = 2 + \varepsilon\tau_1^* + \varepsilon^2\tau_2^*$
Input to $y(\tau^*) = 0$,
Using mathmatica:

$$\begin{cases} \frac{4}{3} - T_1 = 0 \\ \frac{4}{3}T_1 - \frac{T_1^2}{2} - T_2 = 0 \end{cases} \Rightarrow \begin{cases} \tau_1^* = \frac{4}{3} \\ \tau_2^* = \frac{8}{9} \end{cases}$$

$$\tau^* = 2 + \frac{4}{3}\varepsilon + \frac{8}{9}\varepsilon^2 + \cdots$$

Convert back to $t$:

$$t^* = \tau^* \cdot t_c = \frac{V_0}{g}\left(2 + \frac{4}{3}\varepsilon + \frac{8}{9}\varepsilon^2 + \cdots\right)$$

input $\varepsilon = \frac{V_0^2}{gR}$

Consider $O(\varepsilon^1)$:

$$t^* = \frac{V_0}{g}\left(2 + \frac{4}{3}\frac{V_0^2}{gR}\right) = \frac{2V_0}{g} + \frac{4}{3}\frac{V_0^3}{g^2 R}$$

```
In[365]:= Clear[τ, ε, T1, T2]
          清除

(*定义 y*)
y[τ_] := τ - τ^2/2 + ε * (τ^3/3 - τ^4/12);

(*代入 T=T0+ε T1*)
expr = Expand[y[2 + ε T1 + ε^2 T2]];
       展开

(*按 ε 展开到一阶*)
series = Series[expr, {ε, 0, 2}] // Normal
         级数                          转换为普通
```

Out[368]= $\left(\frac{4}{3} - T1\right)\varepsilon + \left(\frac{4\,T1}{3} - \frac{T1^2}{2} - T2\right)\varepsilon^2$

Q3. a) $m\ddot{x} = -kx - c\dot{x}$ , $\alpha = c/\sqrt{mk}$

Conside the natural freq. $\omega_0 = \sqrt{\frac{k}{m}}$ , $[\omega_0] = s^{-1}$

$\therefore$ $t_c = \frac{1}{\omega_0}$ , $\tau = \frac{t}{t_c} = \omega_0 t$

$\Rightarrow \ddot{x} + \frac{c}{m}\dot{x} + \omega_0^2 x = 0$

Let $x_c = \frac{v_0}{\omega_0}$ $\Rightarrow$ $y = \frac{x}{x_c} = \frac{\omega_0}{v_0}x$

$\therefore \frac{dy}{d\tau} = \frac{d(\frac{\omega_0}{v_0}x)}{d(\omega_0 t)} = \frac{1}{v_0}\dot{x}$ $\quad \frac{d^2y}{d\tau^2} = \frac{1}{v_0\omega_0}\ddot{x}$

$\Rightarrow v_0\omega_0\ddot{y} + \frac{c}{m}\cdot v_0\dot{y} + \omega_0^2\frac{v_0}{\omega_0}y = 0$

$\Rightarrow \ddot{y} + \frac{c}{m\omega_0}\dot{y} + y = 0$ $\Rightarrow$ $\ddot{y} + \alpha\dot{y} + y = 0$ $\quad \alpha = \frac{c}{m\omega_0} = \frac{c}{\sqrt{mk}}$

b) $m\ddot{x} + c\dot{x} + kx + Kx^3 = 0$

Consider same thing above : $\omega_0 = \sqrt{\frac{k}{m}}$, and $\varepsilon = K v_0^2 m/k^2$

$t_c = \frac{1}{\omega_0}$ , $\tau = \frac{t}{t_c} = \omega_0 t$

$x_c = \frac{v_0}{\omega_0}$ , $y = \frac{x}{x_c} = \frac{\omega_0}{v_0}x$ $\quad \dot{y} = \frac{1}{v_0}\dot{x}$ , $\ddot{y} = \frac{1}{v_0\omega_0}\ddot{x}$

$\Rightarrow m v_0\omega_0\ddot{y} + c v_0\dot{y} + k\frac{v_0}{\omega_0}y + K\frac{v_0^3}{\omega_0^3}y^3 = 0$

$\Rightarrow \ddot{y} + \frac{c}{m\omega_0}\dot{y} + \frac{k}{m\omega_0^2}y + K\frac{v_0^2}{m\omega_0^4}y^3 = 0$

$\Rightarrow \ddot{y} + \alpha\dot{y} + y + \varepsilon y^3 = 0$

c) For an undamped oscillator : $m\ddot{x} + kx = 0$

The natural freq. $\omega_0 = \sqrt{\frac{k}{m}}$ , $t_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\sqrt{k/m}}$

For a damped time scale : $m\ddot{x} + c\dot{x} = 0$

$x \sim e^{-(c/m)t}$ $\Rightarrow$ $t_d \sim \frac{m}{c}$

the ratio $= \frac{t_0}{t_d} = \frac{2\pi\sqrt{m/k}}{m/c} = 2\pi\frac{c}{\sqrt{mk}} \sim \alpha$

$\alpha$ represent how much damping effect in 1 period.

For spring : $m\ddot{x} + kx + Kx^3 = 0$

$\Rightarrow \ddot{x} + \frac{k}{m}x + \frac{K}{m}x^3 = 0$

The ratio for two kind force: $\frac{F_L}{F_{NL}} = \frac{\frac{k}{m}x}{\frac{K}{m}x^3} = \frac{K}{k}x^2$

Input the nondimensional $x_c = \frac{v_0}{\omega_0}$ , $\frac{F_L}{F_{NL}} = \frac{K}{k}\cdot\frac{v_0^2}{\omega_0^2} = \frac{K}{k}\frac{v_0^2}{k/m} = \varepsilon$

$\therefore$ $\varepsilon$ is the ratio between the nonlinear and linear term of spring

d) $\ddot{y} + \alpha \dot{y} + y + \varepsilon y^3 = 0$

The I.C. $x(0) = 0, \quad \dot{x}(0) = V_0$

$\because \; y = \dfrac{x}{x_c} = \dfrac{\omega_0}{V_0} x \qquad \dot{y} = \dfrac{1}{V_0} \dot{x}$

$\therefore$ I.C.$'$ : $\quad y(0) = 0, \quad \dot{y}(0) = 1.$

Set $\quad y = y_0 + \varepsilon y_1 + O(\varepsilon^2)$

$$\begin{cases} y' = y_0' + \varepsilon y_1' \\ y'' = y_0'' + \varepsilon y_1'' \end{cases}$$

$\Rightarrow \quad y_0'' + \varepsilon y_1'' + \alpha(y_0' + \varepsilon y_1') + y_0 + \varepsilon y_1 + \varepsilon(y_0 + \varepsilon y_1)^2 = 0$

```
In[411]:= Clear[y0, y1, ε, α, t]
          清除
          y = y0[t] + ε y1[t];
          eq = D[y, {t, 2}] + α D[y, t] + y + ε y^3;
               偏导               偏导
          series = Normal[Series[eq, {ε, 0, 2}]]
                   转换⋯  级数
```

Out[414]= $y0[t] + 3\,\varepsilon^2\, y0[t]^2\, y1[t] + \alpha\, y0'[t] + y0''[t] + \varepsilon\,(y0[t]^3 + y1[t] + \alpha\, y1'[t] + y1''[t])$

In[417]:= $y0[t] + 3\,\varepsilon^2\, y0[t]^2\, y1[t] + \alpha\, y0'[t] + y0''[t] + \varepsilon\,(y0[t]^3 + y1[t] + \alpha\, y1'[t] + y1''[t])$

```
          c0 = Coefficient[series, ε, 0]
               系数
          c1 = Coefficient[series, ε, 1]
               系数
          c2 = Coefficient[series, ε, 2]
               系数
```

Out[417]= $y0[t] + 3\,\varepsilon^2\, y0[t]^2\, y1[t] + \alpha\, y0'[t] + y0''[t] + \varepsilon\,(y0[t]^3 + y1[t] + \alpha\, y1'[t] + y1''[t])$

Out[418]= $y0[t] + \alpha\, y0'[t] + y0''[t]$

Out[419]= $y0[t]^3 + y1[t] + \alpha\, y1'[t] + y1''[t]$

Out[420]= $3\, y0[t]^2\, y1[t]$

$$\begin{cases} y_0 + \alpha y_0' + y_0'' = 0 & \qquad y_0(0) = 0, \quad y_0'(0) = 1 \\ y_0^3 + y_1 + \alpha y_1' + y_1'' = 0 & \qquad y_1(0) = 0, \quad y_1'(0) = 1 \\ y_0^2 y_1 = 0 & \end{cases}$$